

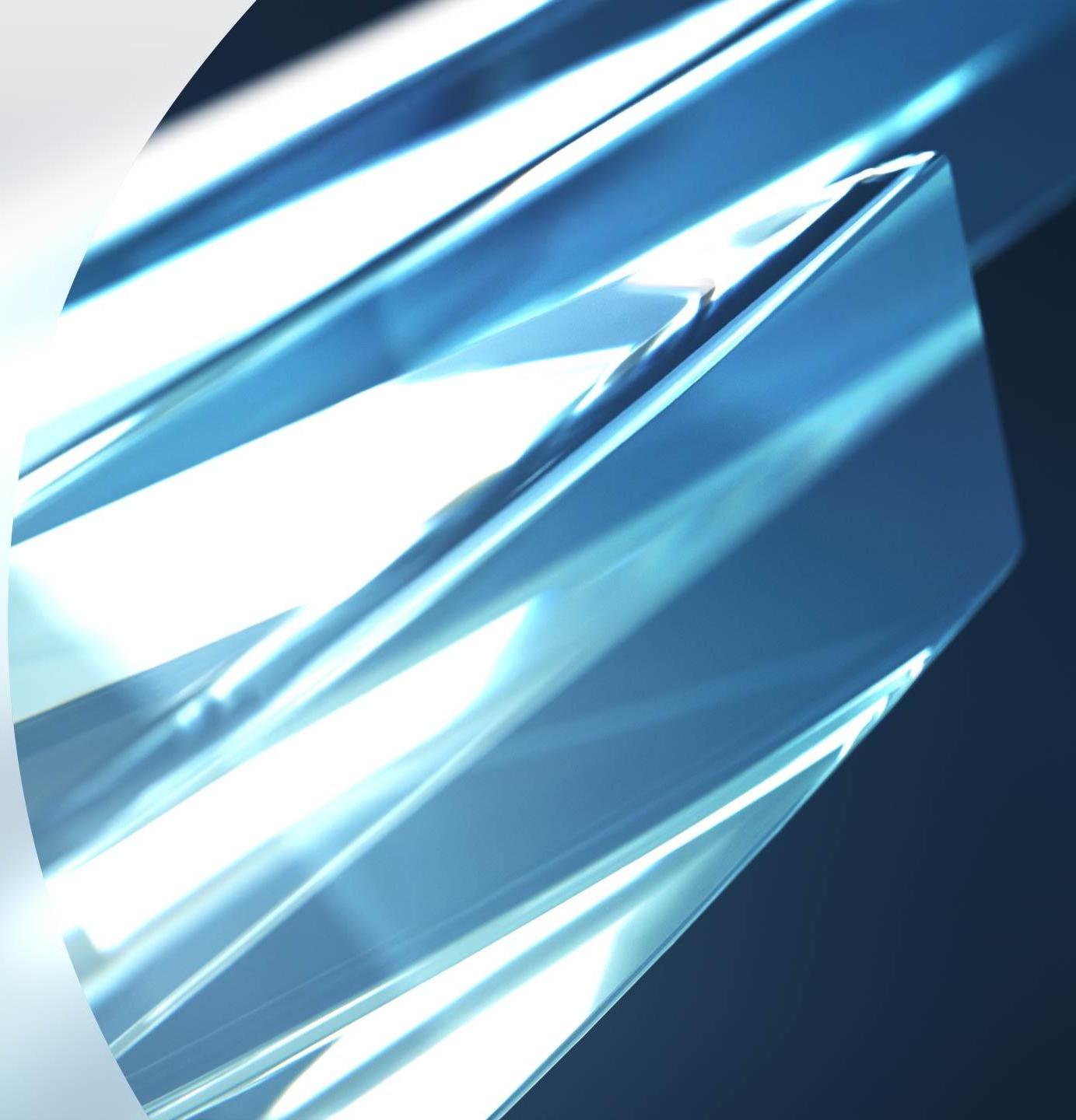


Advancing ITS AMS Through Community Action: A 21st Century Perspective

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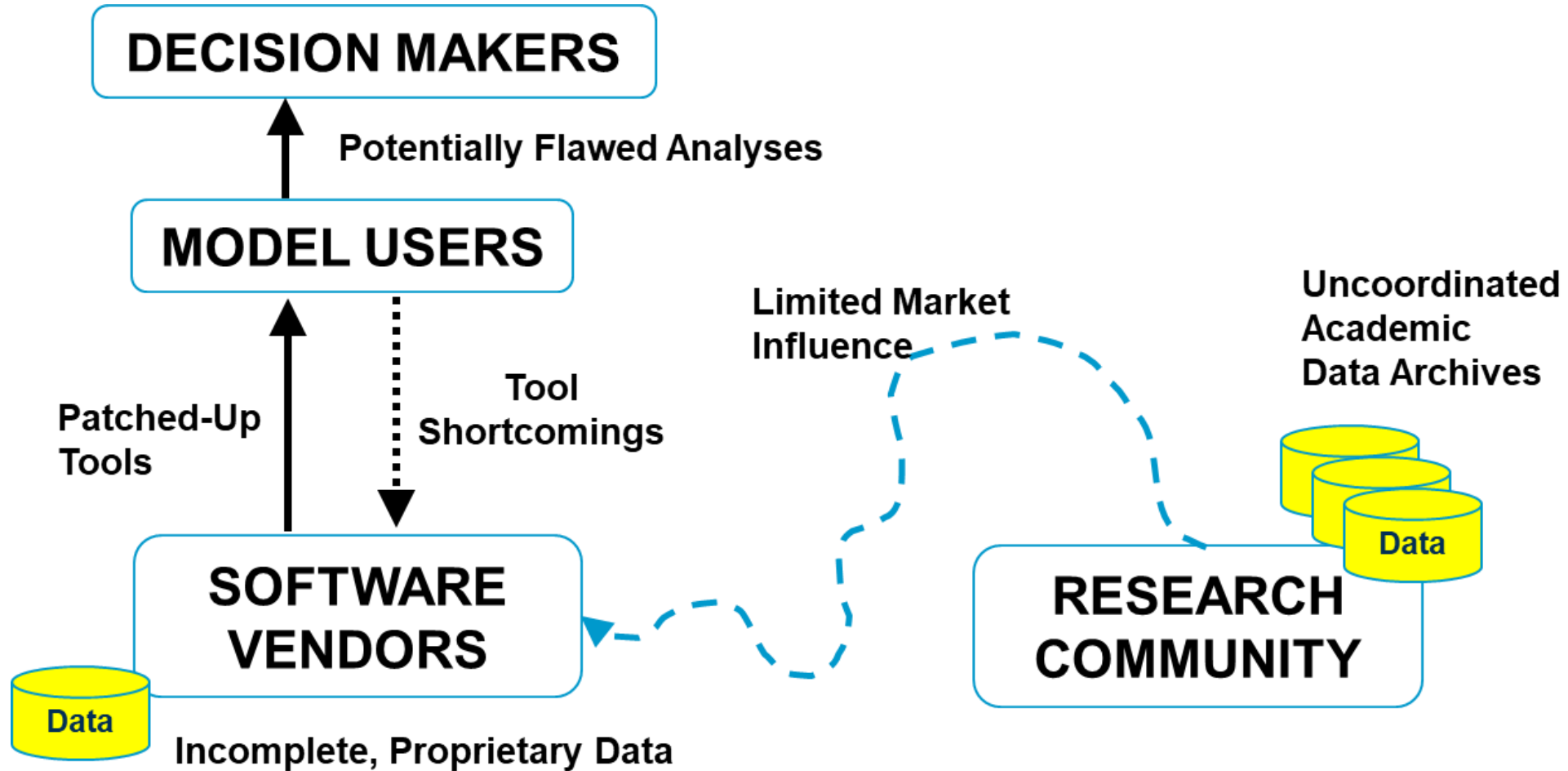


AMS Community Building: Impact (and Implications) of the Next Generation Simulation (NGSIM) Program



- USDOT Program (2001-2008) addressing fundamental issues in the quality of traffic simulation
 - FHWA Office of Operations and Office of Operations R&D and the USDOT ITS JPO
- In the 2000s, the NGSIM program marked a fundamental change in federal role in traffic simulation from *model developer* to *market facilitator*
- Created an NGSIM Community of simulation software vendors, practitioners, and academics
- Decades later, NGSIM program still relevant because of the long-standing impact of the vehicle trajectory data sets collected and shared

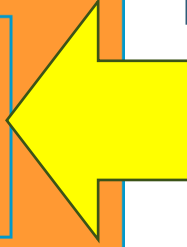
Pre-NGSIM Program ITS AMS Market: Fundamental Flaws



NGSIM Algorithm Prioritization

Rank Most Critical Scenarios

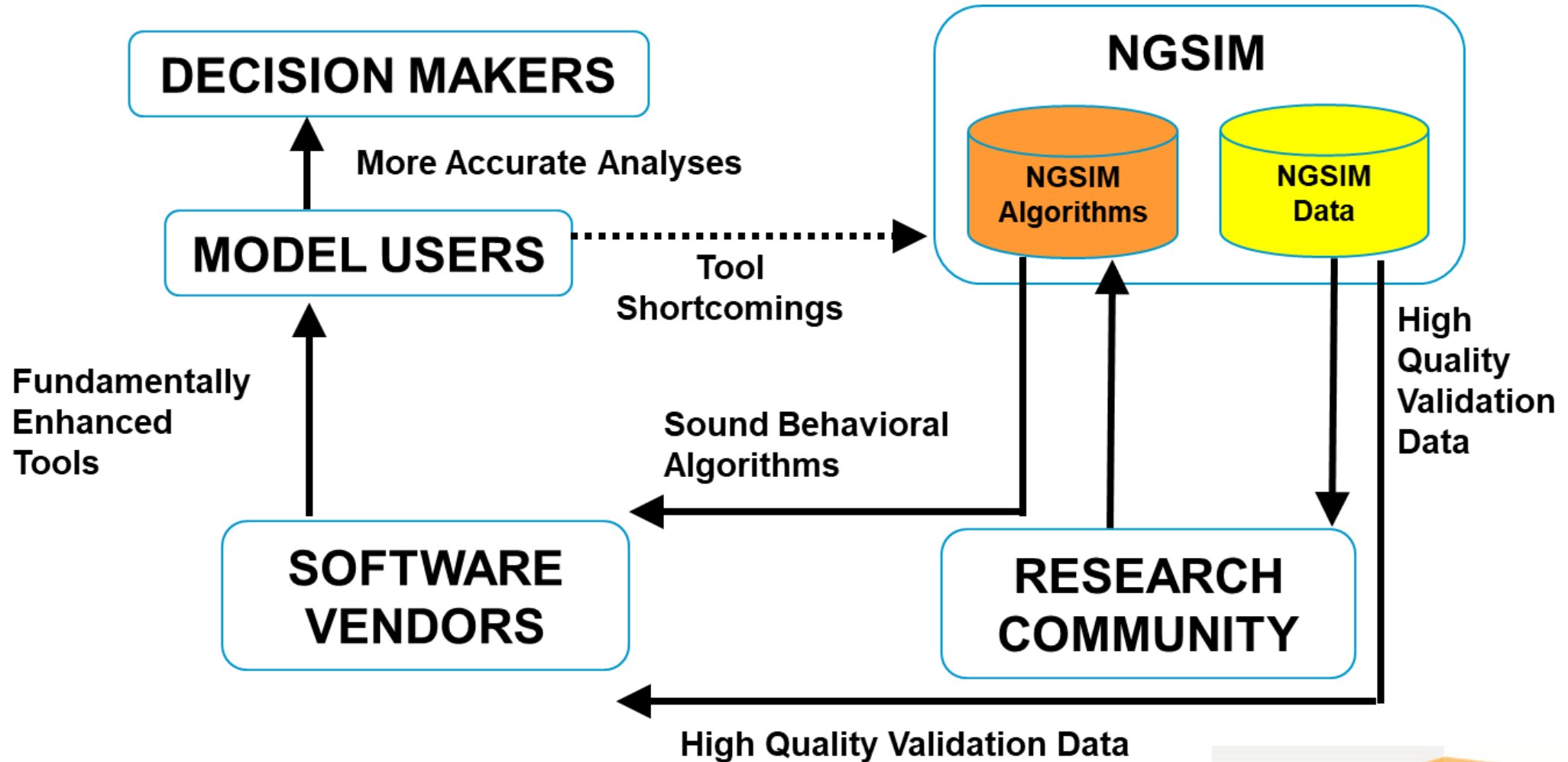
- 1 Lane selection on arterials
- 2 Oversaturated freeways
- 3 Freeway lane distribution
- 4 Weaving sections
- 5 2-way left hand turn lanes
- 6 Response to VMS
- 7 Pedestrians
- 8 Heavy vehicles
- 9 Work zone behavior
- 10 Effects of incidents
- 11 Freeway interchanges
- 12 Urban grid
- 13 Roundabouts
- 14 Freeway geometry
- 15 Transit operations
- 16 Unsafe maneuvers
- 17 Uncontrolled movements
- 18 HOV lane usage
- 19 Illegal Maneuvers
- 20 Stop line behavior



- March 2003 Community Stakeholder Workshop
- 35+ critical scenarios* identified by users where current tools are deficient
 - Poor accuracy under congested conditions was a pervasive concern
- NGSIM program focus: Top 4 critical scenarios
 - Collect comprehensive, all vehicle trajectory data with cameras mounted neighboring tall buildings
 - Key merge/weave and arterial intersections
 - Capture development and dissipation of congestion over 45 minutes

* 35 tactical scenarios plus separate list of strategic traveler choice scenarios

Envisioned NGSIM Impact: Enhanced Tools and Better Decision Support

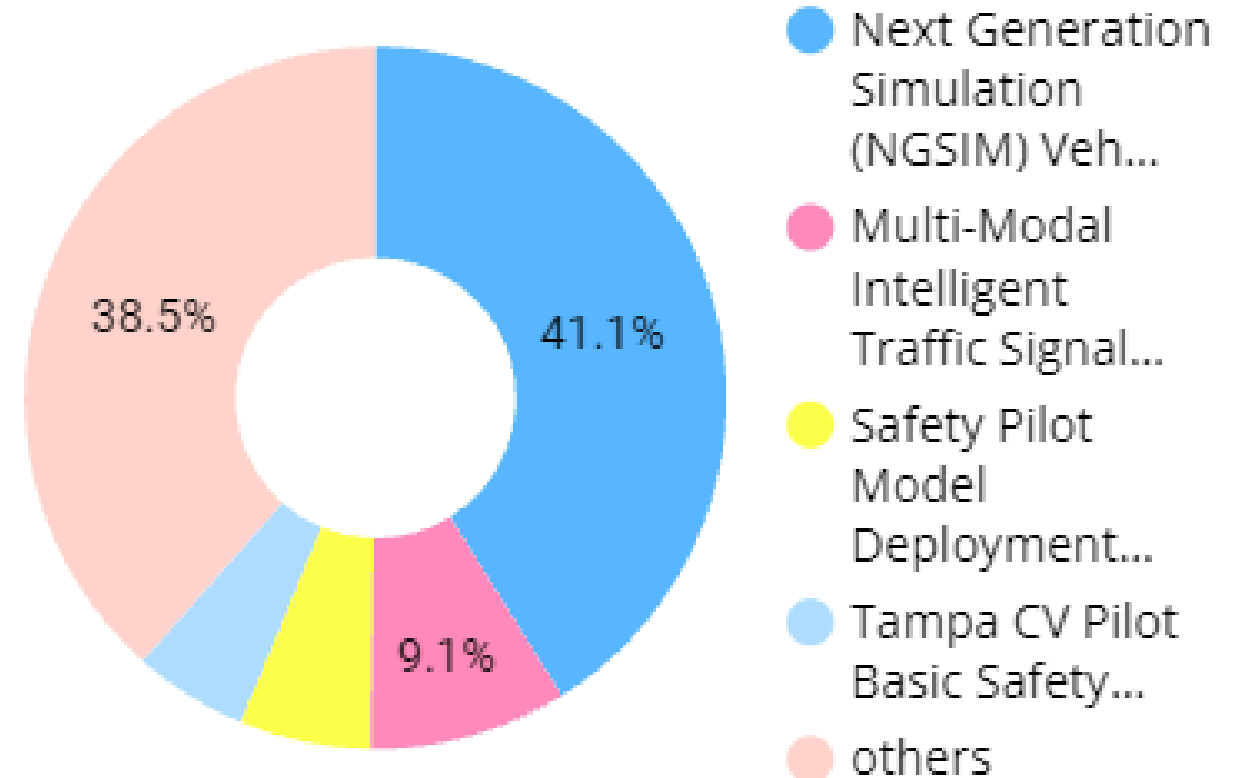


Decades Later: NGSIM Data Sets Continue to Generate Value

- **500+ TRB papers**, conference presentations, and other documents citing NGSIM data
- ITS Data Hub: NGSIM downloaded **more than 140 times** per month
- NGSIM Data Story published on ITS Data Hub

<https://data.transportation.gov/stories/s/i5zb-xe34>

Top DTG Datasets Viewed (all time)

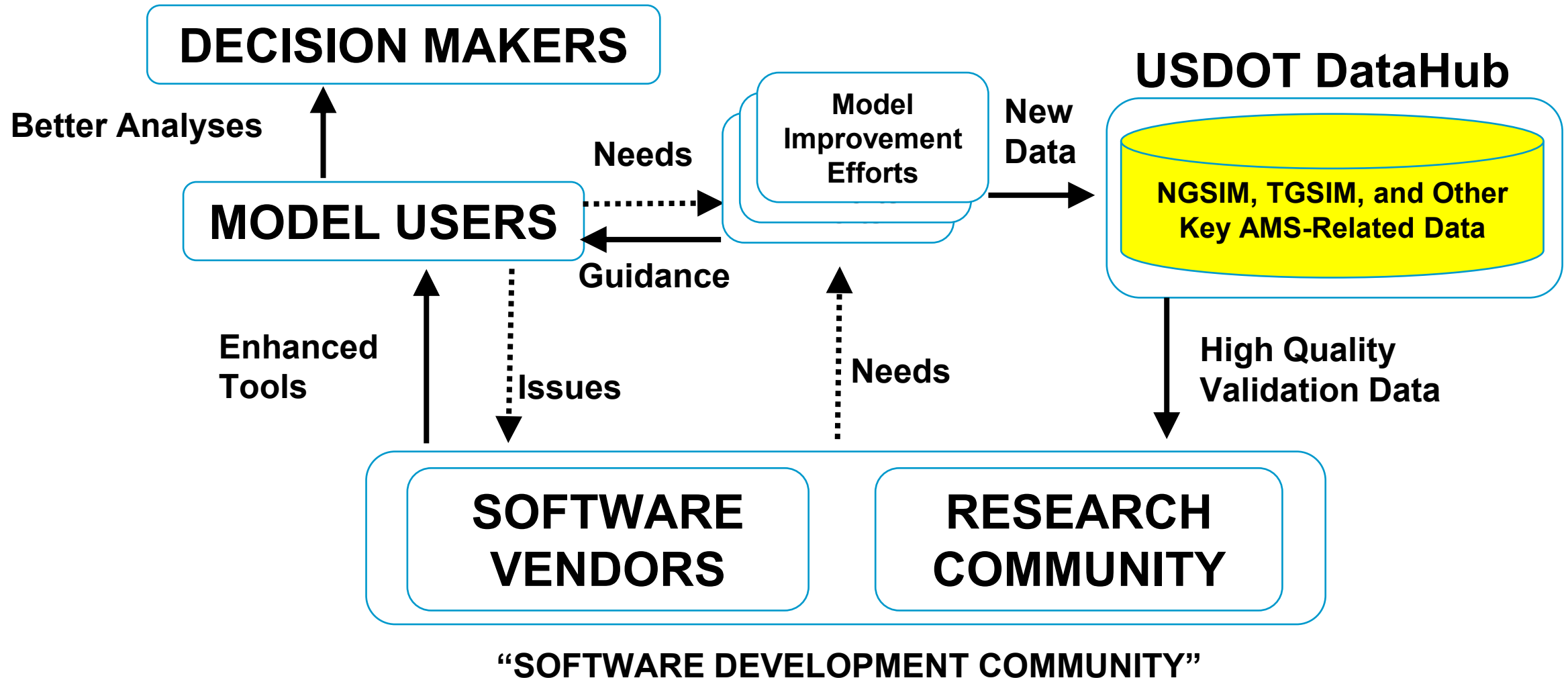


*NGSIM designed as a one-off activity
to change federal role and address 4 top concerns*

22 Years After the NGSIM Community Workshop: What NGSIM Accomplished and What It Did Not

- What NGSIM Accomplished
 - Created ITS AMS community cohesion around four key needed improvements
 - Transformed FHWA/USDOT from market competitor to market facilitator
 - Generated fundamental data sets of lasting value (algorithms/code less durable)
- What NGSIM Did Not Accomplish
 - Did not result in a long-standing, self-organizing ITS AMS community, despite efforts to create one as the NGSIM program wrapped up
 - Did not result in a forum where a new list of prioritized projects/data collection efforts were updated
 - Federal maintenance required to keep data sets and meta-data available
 - Did not inspire others to contribute code or data outside of federally-funded efforts

Revisiting the Post-NGSIM AMS Market: Improved, But Neither Tools Nor Market Are “Perfect”



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2023 CAV/AMS Workshops: Top Actions

Develop Guidance

Modeling, Cost/Benefit, Scenario Planning,
Safety, Calibration, MOEs, Assumptions

Conduct Behavioral Research

Driver, Automated Vehicle, Work Zone
Safety, Fleet Penetration

Establish and Update Standards

Shared Data Format, CAV Definitions,
Cybersecurity

Collect and Disseminate Data

Trajectory, Storage, Standardization,
Lane Change, Car Following

Characterize System Impacts

Penetration, Mixed, Capacity, Demand,
Extreme Conditions, Infrastructure

Map Research Needs to Potential Tools and Workflows

Open Behavior Model, Tool Library

Engage Stakeholders

Industry, Academia, State/Local, Education,
Collaboration, Expert Panels, Facilitation

Research and Develop Use Cases and Pilot Projects

Emerging Tech, Vulnerable Road User
Impacts

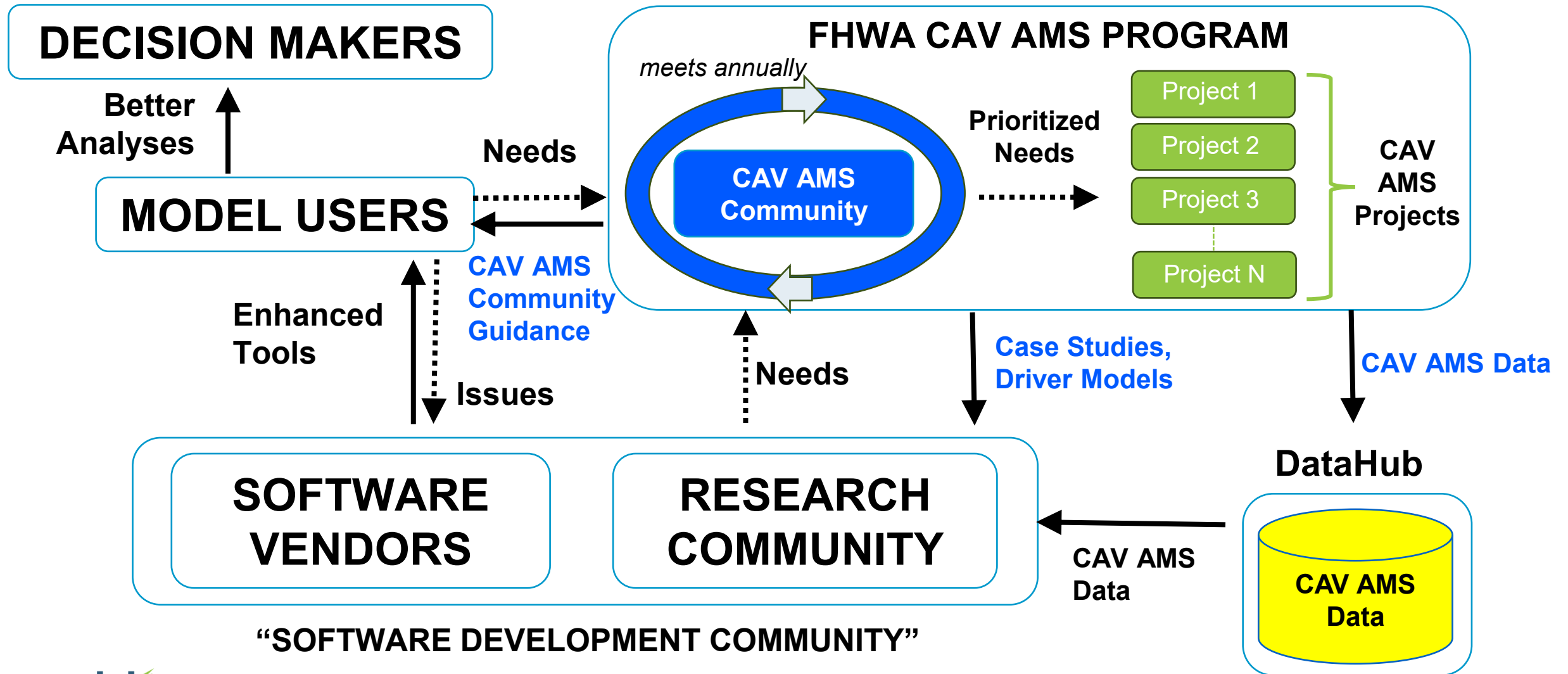
Research Human Factors

Adoption Rates, Decisionmaking

22 Years After the NGSIM Community Workshop: ITS AMS Needs Summary

- What Has Changed
 - More powerful compute platforms, more complex (and diverse) tools, a focus on AI/agent modeling, increased data availability
 - Increased focus on vehicle connectivity and driving automation
 - Complexity of decision-making, difficulty of the ITS AMS “ask”
- What Has Not Changed
 - Desire for more capable yet easier to use tools
 - A desire for “more” data, a focus on understanding and modeling “non-uniform” traveler and driver behavior in response to new technology and societal trends
 - A need to gain decision-maker trust in tools (and assumptions), heightened requirement to uncomplicate the complicated

Hypothesis: An Engaged ITS AMS Community Can Create Sustained Impact on CAV Needs



Benefits of a CAV AMS Community Engagement

- Prioritized assessment of AMS-related needs, updated annually
- Rapid conversion of needs into actionable federal projects
- Venue and mechanism for rapid sharing of CAV AMS guidance and best practices
- Creates data sets of potentially lasting value, which the community can anticipate and rapidly access/utilize
- Results in targeted case studies and behavioral research relevant to the CAV AMS software development community

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